AST5110 Problem 1ex

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April 2026

1 Exercise 1 : Order of Accuracy Analysis when the limit is unknown

In this exercise, we want to determine the order of convergence even when the exact limit is unknown. Instead of comparing with an analytical solution, we use the numerical results computed on three different grids:

$$N, \quad 2N, \quad 4N.$$

2 Convergence model

Assume that the numerical approximation satisfies

$$f_N = f_{\text{lim}} + A\left(\frac{L}{N}\right)^m,$$

where

- f_{lim} is the unknown limiting value,
- A is a constant,
- m is the order of convergence.

Then for the refined grids we have

$$f_{2N} = f_{\text{lim}} + A\left(\frac{L}{2N}\right)^m = f_{\text{lim}} + A\left(\frac{L}{N}\right)^m 2^{-m},$$

and

$$f_{4N} = f_{\text{lim}} + A\left(\frac{L}{4N}\right)^m = f_{\text{lim}} + A\left(\frac{L}{N}\right)^m 4^{-m}.$$

Now subtract:

$$f_{2N} - f_N = A\left(\frac{L}{N}\right)^m (2^{-m} - 1),$$

$$f_{4N} - f_{2N} = A\left(\frac{L}{N}\right)^m (4^{-m} - 2^{-m}).$$

Taking the ratio gives

$$\frac{f_{4N} - f_{2N}}{f_{2N} - f_N} = \frac{4^{-m} - 2^{-m}}{2^{-m} - 1}.$$

Let

$$q = 2^{-m}.$$

Then

$$\frac{4^{-m} - 2^{-m}}{2^{-m} - 1} = \frac{q^2 - q}{q - 1} = q = 2^{-m}.$$

Hence,

$$\frac{f_{4N} - f_{2N}}{f_{2N} - f_N} = 2^{-m}.$$

Taking logarithms, we obtain

$$m = \frac{\log\left|\dfrac{f_{2N} - f_N}{f_{4N} - f_{2N}}\right|}{\log 2}.$$

Therefore, the convergence order can be computed using only numerical values on three grids, without knowing the exact limit.

3 Test of `deriv_cent`

For the standard central difference approximation of the first derivative,

$$D_c f(x) = \frac{f(x+h) - f(x-h)}{2h},$$

we use Taylor expansion:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f^{(3)}(x) + O(h^4),$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f^{(3)}(x) + O(h^4).$$

Subtracting these two expressions gives

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3}f^{(3)}(x) + O(h^5).$$

Dividing by $2h$, we get

$$D_c f(x) = f'(x) + \frac{h^2}{6}f^{(3)}(x) + O(h^4).$$

So the leading error term is of order $O(h^2)$, and therefore

$$\boxed{m = 2.}$$

Thus, `deriv_cent` is second-order accurate.

3.1 Test of `deriv_4tho`

For the fourth-order finite difference approximation,

$$D_4 f(x) = \frac{-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)}{12h},$$

the Taylor expansion shows that

$$D_4 f(x) = f'(x) - \frac{h^4}{30} f^{(5)}(x) + O(h^6).$$

Hence the leading error is of order $O(h^4)$, so

$$\boxed{m = 4.}$$

Thus, `deriv_4tho` is fourth-order accurate.

3.2 Implementation of `order_conv`

The function `order_conv` should take the three numerical values

$$f_N, \quad f_{2N}, \quad f_{4N}$$

as input and return the estimated convergence order

$$\boxed{m = \frac{\log \left| \dfrac{f_{2N} - f_N}{f_{4N} - f_{2N}} \right|}{\log 2}}.$$

3.3 Conclusion

Even when the exact analytical limit is unknown, we can still estimate the order of convergence by comparing numerical results on three grids. For the two derivative schemes considered here, we obtain:

$$\boxed{\texttt{deriv\_cent} \text{ is second-order accurate}},$$

$$\boxed{\texttt{deriv\_4tho} \text{ is fourth-order accurate}}.$$